PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Prognostic role of tumor-infiltrating lymphocytes assessed by hematoxylin and eosin-stained section in gastric cancer: A systematic review and meta-analysis
AUTHORS	Tian, Chunfang; Jing, haiyan; Wang, Caixia; Wang, Weibo; Cui, Yangang; Chen, Jianpeng; Sha, Dan

VERSION 1 – REVIEW

REVIEWER	Arpad V. Patai 1st Department of Surgery and Interventional Gastroenterology, Semmelweis University, Hungary
REVIEW RETURNED	15-Oct-2020

GENERAL COMMENTS	Tian et al. performed a systematic review and meta-analysis on the prognostic role of TILs by HE sections in gastric cancer. This a well-performed study, but the manuscript needs some improvement before it can be accepted for publication. Major comments From the 9 included studies, 8 were conducted in Asia. 1. Does this influence study results? 2. Is there a difference in TIL density in (any) tumors between East vs West? 3. Is there a difference in demographic factors, histologic subtypes, tumorigenesis (including etiological and risk factors), molecular subtype distribution, stage at diagnosis and clinical outcome in GC between East and West? 4. How was TIL density evaluated in the included studies (manually or digitally)? Was there a difference in methodology? 5. Different cut-offs in TIL density might affect the results, please comment on this. 6. What cut-off for TIL categories would the authors suggest for clinical practice? Please comment on this in the discussion. 7. The authors detected publication bias in histological grades and Lauren classification. Could you please specify that? This should be discussed, and also mentioned in limitations. 8. Is there any data on the efficacy and adverse event of chemotherapy based on TIL status or other molecular subtypes (MSI, EBV, CIN etc.) in GC? Please comment on this in the discussion. Minor comments

	1. Electronic links should be provided in Supplementary data 2. Manually checked reference list is not provided in the flow chart Typos Eg.: p2 line 46 TIL density was p3 line 7 fibroblasts p7 line 4 fixed effects model Language The MS needs some language polishing.

REVIEWER	Hee Eun Lee Mayo Clinic Rochester, MN, USA
REVIEW RETURNED	20-Oct-2020

GENERAL COMMENTS	It has been shown that tumor-infiltrating lymphocytes (TILs) have clinical significance by playing a role in prognostication and prediction of treatment response to immune checkpoint inhibitors in gastric cancers. In this manuscript, the authors suggested TIL density manually counted on H and E slides is a reliable biomarker to predict clinical outcome of gastric cancer patients by showing that it was significantly associated with TNM stage and patient's overall survival as TIL density counted on immunohistochemically stained slides are, by meta-analysis. Their methods appear to be reasonable and scientific, and the results are clinically meaningful as showing a possibility that TIL density counted on routine histologic examination (H and E sections) can be used in clinical practice. I have a few things to address. 1. TIL density manually counted on HE sections is generally less reproducible than that counted using Immunohistochemical and/or digital image analysis method. So please delete "reproducible" page 3 line 9 and page 13 line 12. 2. The clinical significance of intratumoral versus stromal TILs is not well established so far in gastric caners. I recommend adding more thorough review of literature in discussion. 3. Page 13 line 33-41, "Compared with ….. TIL evaluation." --- I do not think this sentence is true. One shouldn't generalize stromal TILs are more reproducible. Intratumoral TILs and stromal TILs both often show heterogeneity within a tumor. Also, intratumoral TILs are not particularly more difficult to count than stromal TILs. Please delete the sentence above.

REVIEWER	Marc Saez University of Girona, Spain
REVIEW RETURNED	25-Oct-2020

GENERAL COMMENTS	The authors attempted to evaluate the association between H & E-based TILs density and prognosis, and clinicopathological characteristics in patients with gastric cancer. The authors have been quite successful in achieving their objective. However, I still have some comments. 1.- The authors say they perform a fixed-effects meta-analysis when heterogeneity was high. They consider that heterogeneity was high when $p < 0.1$. However, the number of studies to be meta-analyzed is very small, at most 9. Therefore, firstly, they should increase the risk much more (from 10% to 15%, even 20%)

	and, secondly, the forest plots indicate that the heterogeneity is very high, even if it has not been detected . They should always use random effects. 2.- The authors say they find publication bias in the histological grade and Lauren classification. However, they do not correct it. They should do so or explain, in detail, its consequences.

REVIEWER	Luis Puente Maestu Hospital Universitario Gregorio Marañón- Facultad de Medicina de la Universidad Complutense de Madrid. Spain
REVIEW RETURNED	25-Oct-2020

GENERAL COMMENTS	Page 8,line 7 The Egger's tests and Begg's tests were conducted to detect publication bias. A reference to theses test is needed

REVIEWER	Moritz Tobiasch Landeskrankenhaus Hall in Tirol, Austria
REVIEW RETURNED	01-Nov-2020

GENERAL COMMENTS	A straightforward, concice study on tumor infiltrating lymphocyte, using a standard histopathologic measures. There is in my view only one issue to state: when choosing whether a random- or fixed effects model is to be used, one must not only take into consideration if outcome parameters are sufficiently comparable (as did the authors), but also estimate the influence of possible unaccounted confounders. In general, as data reporting in primary research reports tends to be of variable quality (as also mentioned by the authors in this case) it might be advisable to calculate, in meta-analytic biomedical research, all effects on the basis of random-effects models. I would therefore suggest to re-run the fixed-effect calculations and compare the results. Most likely, this will only affect confidence intervals and much less the effect estimate itself. Another minor comment: as shown here (in reproduction of known data) tumors high in TIL are less advanced at the timepoint of diagnosis and seem to have a quite fundamentally different tumor biology. Maybe this fact could be discussed a little more in detail. Apart from that, I may congratulate the authors for this study.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Reviewer Name: Arpad V. Patai

Comments to the Author

Comment:
From the 9 included studies, 8 were conducted in Asia.
1. Does this influence study results?

Thank you for your careful and high-quality review, Reviewer 1!

Answer:
We agree with “eight of the nine included studies were conducted in Asia”. According to the “Global Cancer Statistics 2018: GLOBOCAN Estimates of Incidence and Mortality Worldwide for 36 Cancers in 185 Countries (Bray F, Ferlay J, Soerjomataram I, et al. Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA Cancer J Clin 2018;68(6):394-424.)”, almost two-third of gastric cancers occur in Asia. Republic of Korea, Japan and China are the countries with the highest incidence rates of gastric cancer worldwide. Most studies included in our meta-analysis come from Republic of Korea, Japan and China and represent the geographic distribution of gastric cancer.

Comment:
2. Is there a difference in TIL density in (any) tumors between East vs West?

Answer:
Yes. There is a difference in TIL density in gastric cancers (GCs) between East vs. West. One study (Lin SJ, Gagnon-Bartsch JA, Tan IB, et al. Signatures of tumor immunity distinguish Asian and non-Asian gastric adenocarcinomas. Gut 2015; 64(11):1721-31) compared gene expression profiles of GCs (n=1016) between six Asian cohorts (2 from Singapore, 1 from Hong Kong, 3 from South Korea) and three non-Asian cohorts (1 from Australia, 1 from Netherlands, 1 from the United Kingdom). Non-Asian GCs showed significantly higher expression of T-cell markers (CD3, CD45R0, CD8) and lower expression of the immunosuppressive T-regulatory cell marker FOXP3 compared to Asian GCs. Inflammatory cell markers CD66b and CD68 also exhibited significant cohort differences.

Comment:
3. Is there a difference in demographic factors, histologic subtypes, tumorigenesis (including etiological and risk factors), molecular subtype distribution, stage at diagnosis and clinical outcome in GC between East and West?

Answer:
There are some differences in demographic factors, histologic subtypes, tumorigenesis (including etiological and risk factors), molecular subtype distribution, stage at diagnosis and clinical outcome in GC between East and West.
1). Demographic factors:
According to the “Global Cancer Statistics 2018 (Bray F, Ferlay J, Soerjomataram I, et al. Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA Cancer J Clin 2018;68(6):394-424.)”, almost two-third of gastric cancers occurs in Asia. The Republic of Korea, Japan and China are the countries with the highest incidence rates of gastric cancer worldwide. The incidence rates of gastric cancer are markedly elevated in East, whereas the rates in West are generally low.

2). Histologic subtypes
Intestinal type classification predominated in both Asian and Caucasian, but there were more American patients with a diffuse classification than in Chinese patients. (Strong VE, Wu AW, Selby LV, et al. Differences in gastric cancer survival between the U.S. and China. Journal of surgical oncology 2015;112(1):31-7). The Eastern patients had a higher percentage of differentiated tumors compared to Western patients. (Lin JX, Lin JP, Desiderio J, et al. Difference in the short-term outcomes of laparoscopic gastrectomy for gastric carcinoma between the east and west: a retrospective study from the IMIGASTRIC trial. Journal of Cancer 2019;10(17):4106-13).

3). Tumorigenesis (including etiological and risk factors)

The epidemiologic and risk factors, including Helicobacter pylori, diet or smoking, are universal risk factors for gastric cancer, irrespective of geography. (Suh YS, Yang HK. Screening and Early Detection of Gastric Cancer: East Versus West. Surg Clin North Am 2015;95(5):1053-66).

4). Molecular subtype distribution
Recent comprehensive molecular profiling of gastric cancer proposed four molecular subtypes, i.e. Epstein-Barr virus-associated, microsatellite instable, chromosomal instable and genomically stable carcinomas. The authors did not observe any systematic differences in distribution of subtypes between patients of East Asian and Western origin (Cancer Genome Atlas Research N. Comprehensive molecular characterization of gastric adenocarcinoma. Nature 2014;513(7517):202-9.). Another study found that Hispanic/Latino patients had a significantly larger proportion of genomically stable subtype tumors compared to Asian and White patients. (Wang SC, Yeu Y, Hammer STG, et al. Hispanic/Latino Patients with Gastric Adenocarcinoma Have Distinct Molecular Profiles Including a High Rate of Germline CDH1 Variants. Cancer research 2020;80(11):2114-24.)

5).Stage at diagnosis
In Western countries, most patients are diagnosed with advanced and unresectable gastric cancers. However, in some Eastern countries, such as South Korea and Japan, an established national screening program permits frequent diagnosis of early-stage disease. (Suh YS, Yang HK. Screening and Early Detection of Gastric Cancer: East Versus West. Surg Clin North Am 2015;95(5):1053-66).

6). Clinical outcome
An analysis of the effect of race on the prognosis after curative gastrectomy revealed that Asian race was an independent predictor of overall survival which provided a better prognosis. On the contrary, Caucasian race provided a worse overall prognosis. (Griniatsos J, Trafalis D. Differences in gastric cancer surgery outcome between East and West: differences in surgery or different diseases? Journal of BUON : official journal of the Balkan Union of Oncology 2018;23(5):1210-15) .

Comment:
4. How was TIL density evaluated in the included studies (manually or digitally)? Was there a difference in methodology?

Answer:
TIL density were evaluated manually in 8 of the 9 included studies, and was evaluated digitally in only one study. Four studies evaluated TILs based on the International Immuno-Oncology Biomarkers Working Group (IIOBWG) recommendations, and other five studies evaluated TILs based on different approaches.

Comment:
5. Different cut-offs in TIL density might affect the results, please comment on this.

Answer:
We revised the part of “the limitations in our study” as follows: “the methodology for evaluating TILs and its cutoff value varied across different studies, which might affect the results of our meta-analysis.”

Comment:
6. What cut-off for TIL categories would the authors suggest for clinical practice? Please comment on this in the discussion.

Answer:

To date, there is no recommendation on TIL cutoff for any tumor type. We mentioned this at the end of the third paragraph in the discussion part: “Despite no consensus for TIL evaluation and its cutoff in gastric cancer, our meta-analysis suggests TIL density evaluated on H&E stained sections is a reliable tool to predict survival in this cancer type. Future study is needed to explore the optimal evaluation method and cutoff value for TILs”.

Comment:
7. The authors detected publication bias in histological grades and Lauren classification. Could you please specify that? This should be discussed, and also mentioned in limitations.

Answer:
We adjusted the publication bias and mentioned it in the method and result parts. We added the following statement in the Statistical analysis of the method: “The trim and fill method was used to evaluate the effect of publication bias on the pooled results.”
We added the following statement in the Publication bias and sensitivity analysis of the results: “The trim and fill method indicated that studies were unlikely to be missing in the analysis of the association between TILs and histological grade, while two studies were found missing in the analysis of the association between TILs and Lauren classification (figure 4). The effects of TILs in histological grade and Lauren classification were stable and not changed after having adjusted the publication bias.”
We added the following statement in the part of limitations: “In addition, publication bias in histological grades and Lauren classification was detected and adjusted by the trim and fill method in our meta-analysis.”

Comment:
8. Is there any data on the efficacy and adverse event of chemotherapy based on TIL status or other molecular subtypes (MSI, EBV, CIN etc.) in GC? Please comment on this in the discussion.

Answer:
Yes! We have added the following statement at the second paragraph of the Discussion: “High TILs were also associated with the efficacy of chemotherapy in gastric cancer. Li et al. found that high $CD3^+$ or $FoxP3^+$ T-cell infiltration was associated with better overall survival in patients with stage II–III gastric cancer treated with FOLFOX (Li W, Wang W, Liao P, et al. High levels of tumor-infiltrating lymphocytes showed better clinical outcomes in FOLFOX-treated gastric cancer patients. Pharmacogenomics 2020;21(11):751-59).”

Minor comments
1. Electronic links should be provided in Supplementary data

Answer:
We provided the electronic links to full PubMed search strategy in Supplementary data and referred to it in the methods section. We added the following statement in the part of search strategy: “The full electronic search strategy for PubMed can be accessed in the Supplementary File (supplementary tables S1-S2).”

Minor comments
2. Manually checked reference list is not provided in the flow chart

Answer:
We have added the manually checked reference list in the flow chart.

Typos
Eg.:

p2 line 46 TIL density was
p3 line 7 fibroblasts
p7 line 4 fixed effects model

Answer:
We have revised the sentences mentioned above.

Language
The MS needs some language polishing.

Answer:
We asked an English native speaker to polish language in our manuscript.

Reviewer: 2
Reviewer Name: Hee Eun Lee

Comments to the Author

Comment:
1. TIL density manually counted on HE sections is generally less reproducible than that counted using Immunohistochemical and/or digital image analysis method. So please delete “reproducible” page 3 line 9 and page 13 line 12.

Thank you for your careful and high-quality review, Reviewer 2!
Answer:
We followed Reviewer 2’s suggestion and deleted “reproducible” in the sentences mentioned above.

Comment:
2. The clinical significance of intratumoral versus stromal TILs is not well established so far in gastric caners. I recommend adding more thorough review of literature in discussion.

Answer:
We revised our paper (in the discussion part, the 4th paragraph) as follows: “To date, prognostic value of TIL’s locations vary in different reports and there are no recommendations for evaluation of TILs in gastric cancer. Our meta-analysis showed high TILs on H&E slides are associated with better OS independent of TIL locations, suggesting intratumoral TILs, stromal TILs, and total TILs might be reliable prognostic biomarkers for gastric cancer. A meta-analysis reported that high density of CD3+ TILs evaluated in all areas, including cancer nest and cancer stroma, was associated with longer OS in gastric cancer. One study indicated high intratumoral TILs were related to prolonged OS in patients with gastric adenocarcinoma. On the contrary, in another study including 120 patients with Epstein-Barr virus (EBV)-positive gastric adenocarcinoma, the prognostic value was only seen in stromal TILs, but not in intratumoral TILs. EBV-positive gastric cancers displayed gene mutation (PIK3CA), gene amplification (JAK2, CD274 and PDCD1LG2) and DNA hypermethylation. The association of TIL locations with molecular features awaits further study. Future research that explores the biological differences of TILs in different locations is needed to better understand the host antitumor immune responses.”

Comment:
3. Page 13 line 33-41, “Compared with ….. TIL evaluation.” --- I do not think this sentence is true. One shouldn’t generalize stromal TILs are more reproducible. Intratumoral TILs and stromal TILs

both often show heterogeneity within a tumor. Also, intratumoral TILs are not particularly more difficult to count than stromal TILs. Please delete the sentence above.

Answer:
We followed Reviewer 2's suggestions and deleted the sentences mentioned above.

Reviewer: 3
Reviewer Name: Marc Saez

Comments to the Author

Comment:
1. - The authors say they perform a fixed-effects meta-analysis when heterogeneity was high. They consider that heterogeneity was high when p <0.1. However, the number of studies to be meta-analyzed is very small, at most 9. Therefore, firstly, they should increase the risk much more (from 10% to 15%, even 20%) and, secondly, the forest plots indicate that the heterogeneity is very high, even if it has not been detected. They should always use random effects.

Thank you for your careful and high-quality review, Reviewer 3!
Answer:
Thank you for your professional suggestions. We increased the risk in the part of Statistical analysis (heterogeneity was considered high for $P < 0.15$ and $I^2 > 50\%$) and used the random-effect model to re-run the fixed effects calculations.

Comment:
2. - The authors say they find publication bias in the histological grade and Lauren classification. However, they do not correct it. They should do so or explain, in detail, its consequences.

Answer:
We adjusted the publication bias. We added the following statement in the Statistical analysis of the methods part: "The trim and fill method was used to evaluate the effect of publication bias on the pooled results."
We added the following statement in the Publication bias and sensitivity analysis of the results: "The trim and fill method indicated that studies were unlikely to be missing in the analysis of the association between TILs and histological grade, while two studies were found missing in the analysis of the association between TILs and Lauren classification (figure 4). The effects of TILs in histological grade and Lauren classification were stable and not changed after having adjusted the publication bias. "
We added the following statement in the part of limitations: "In addition, publication bias in histological grades and Lauren classification was detected and adjusted by the trim and fill method in our meta-analysis."

Reviewer: 4
Reviewer Name: Luis Puente Maestu

Comments to the Author

Comment:

Page 8, line 7 The Egger's tests and Begg's tests were conducted to detect publication bias. A reference to theses test is needed

Answer:
We added the references to the Egger's tests and Begg's tests.

Reviewer: 5
Reviewer Name: Moritz Tobiasch

Comments to the Author

Comment:
A straightforward, concise study on tumor infiltrating lymphocyte, using a standard histopathologic measures. There is in my view only one issue to state: when choosing whether a random- or fixed effects model is to be used, one must not only take into consideration if outcome parameters are sufficiently comparable (as did the authors), but also estimate the influence of possible unaccounted confounders. In general, as data reporting in primary research reports tends to be of variable quality (as also mentioned by the authors in this case) it might be advisable to calculate, in meta-analytic biomedical research, all effects on the basis of random-effects models. I would therefore suggest to re-run the fixed-effect calculations and compare the results. Most likely, this will only affect confidence intervals and much less the effect estimate itself.

Thank you for your careful and high-quality review, Reviewer 5!
Answer:
Thank you for your professional suggestions. We used the random-effect model to re-run the fixed-effect calculations. You are right: the re-calculations only affected confidence intervals and did not change the effect estimate itself.

Comment:
Another minor comment: as shown here (in reproduction of known data) tumors high in TIL are less advanced at the timepoint of diagnosis and seem to have a quite fundamentally different tumor biology. Maybe this fact could be discussed a little more in detail.

Answer:
We added the following sentences in the discussion part, 2nd paragraph: "Our previous study in colon cancer (n=1532) indicated tumors with high TILs had distinct tumor biology, were more likely to present with a lower T and N stage , well or moderate histologic grade, right-sided primary location, mutated BRAF $^{V\,600E}$, and dMMR. (Lee H, Sha D, Foster NR, et al. Analysis of tumor microenvironmental features to refine prognosis by T, N risk group in patients with stage III colon cancer (NCCTG N0147) (Alliance). Annals of oncology 2020, 31(4): 487-494.). Patients with high TILs were more likely to achieve response to ICIs, while low TILs represent existing but partially exhausted antitumor immunity. (Dieci MV, Radosevic-Robin N, Fineberg S, et al. Update on tumor-infiltrating lymphocytes (TILs) in breast cancer, including recommendations to assess TILs in residual disease after neoadjuvant therapy and in carcinoma in situ: A report of the International Immuno-Oncology Biomarker Working Group on Breast Cancer. Seminars in cancer biology 2018;52(Pt 2):16-25. Loupakis F, Depetris I, Biason P, et al. Prediction of Benefit from Checkpoint Inhibitors in Mismatch Repair Deficient Metastatic Colorectal Cancer: Role of Tumor Infiltrating Lymphocytes. The oncologist 2020;25(6):481-87.) High TILs were also associated with the efficacy of chemotherapy in gastric cancer. Li et al. found that high CD3+ or FoxP3+ T-cell infiltration was associated with better

overall survival in patients with stage II–III gastric cancer treated with FOLFOX (Li W, Wang W, Liao P, et al. High levels of tumor-infiltrating lymphocytes showed better clinical outcomes in FOLFOX-treated gastric cancer patients. Pharmacogenomics 2020;21(11):751-59).

VERSION 2 – REVIEW

REVIEWER	Arpad V. Patai, MD, PhD Semmelweis University, Hungary
REVIEW RETURNED	07-Dec-2020

GENERAL COMMENTS	The authors answered all my questions and made the required amendments.

REVIEWER	Hee Eun Lee Mayo Clinic, Rochester, MN, USA
REVIEW RETURNED	28-Dec-2020

GENERAL COMMENTS	The manuscript is much improved after revision. I do not have further comments.

REVIEWER	Marc Saez University of Girona, Spain
REVIEW RETURNED	20-Dec-2020

GENERAL COMMENTS	The authors have responded very well not only to my comments but also to those of the other reviewers. In addition, they have incorporated a large part of them in the new version of the manuscript. I have no further comments.

REVIEWER	Moritz Tobiasch LKH Hall in Tirol, Austria
REVIEW RETURNED	14-Dec-2020

GENERAL COMMENTS	In my humble opinion all issues raised by the reviewers were addressed. I would like to suggest to accept this paper for publication.